When the Economy Turns Grey, So Does Fashion

A Machine Learning Analysis of Color Trends and Macroeconomic Conditions

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2026-03-31

Executive Summary

Fashion consumer data has long served as a barometer of economic sentiment. The so-called *lipstick effect*—consumers trading down to small, affordable luxuries in downturns—is well documented in industry and financial circles (JPMorgan Chase 2023). Yet a subtler pattern has been identified by consumers: the apparent drift of runway aesthetics toward muted, neutral palettes coinciding with periods of economic uncertainty (Fashion Times 2023).

This project takes a more systematic approach to see if this trend identified online can be identified in high fashion imagery. Using computer vision and unsupervised machine learning, I analyze **2,400 runway images** from the *Vogue Runway* archive (2000–2023) to quantify shifts in color saturation over time and test whether those shifts correlate with macroeconomic conditions.

This expanded analysis incorporates **new data sources** — Google Trends search interest data, the University of Michigan Consumer Sentiment Index, GDP growth, and S&P 500 returns — alongside **market context** from the emerging "quiet luxury" movement and luxury peer performance data (Accio Consumer Insights 2024; McKinsey & Company 2026).

Central question: when the economic outlook becomes less bright, does fashion follow suit?

Introduction

Research Question

> *Does fashion become more visually neutral when economic conditions worsen?*

Hypothesis

Worsening economic conditions are associated with **lower average chroma** in runway fashion imagery. In practical terms, periods of higher unemployment or recession correspond with more muted, desaturated color palettes—think greys, blacks, and earth tones—while economic expansions correspond with more vivid, saturated looks.

Data & Measures

The analysis combines multiple data sources:

	Description
Fashion imagery	*Vogue Runway* dataset via the Internet Archive; 100 runway looks sampled per year, 2000–2023
Macroeconomic data	Annual U.S. unemployment rates and NBER recession indicators (Federal Reserve Bank of St. Louis 2026a)
Consumer sentiment	University of Michigan Consumer Sentiment Index (UMCSENT) (Federal Reserve Bank of St. Louis 2026b)
Search interest	Google Trends data for "quiet luxury," "neutral fashion," and related terms (Google 2026)
Market context	Quiet luxury market sizing and luxury brand peer performance (Accio Consumer Insights 2024)

The **dependent variable** is *mean chroma*: a measure of how vivid or saturated the average fashion palette is in a given year, extracted via unsupervised machine learning on runway images.

Independent variables include:

- Annual unemployment rate

- Consumer sentiment index
- Recession indicator (binary)
- GDP growth rate
- S&P 500 annual returns
- Lagged economic indicators (prior-year conditions)
- Year (as a control for long-run trend)

Constructing a Measure of Neutrality

Image Processing Pipeline

Before extracting color information, I applied **background removal** to isolate clothing from distracting runway elements—lighting rigs, backdrops, audience. I used Python's `rembg` library, which applies deep learning–based segmentation to retain only the foreground subject.

Each image was then processed using **k-means clustering** ($k = 8$) to extract its eight dominant palette colors. This unsupervised method groups similar pixels by color similarity, giving a compact representation of each look's visual palette (Zou et al. 2016).

Color Metrics

From those extracted palettes, I construct two complementary metrics:

> **Mean chroma** — the average color intensity across a look's palette. Lower values = more muted, desaturated tones. Higher values = more vivid, saturated colors.
> **Neutral share** — the proportion of palette colors falling below a specified chroma threshold. Captures *dominance* of neutral tones, rather than simply averaging intensity.

Results are aggregated by year.

Results

Color Trends Over Time

As expected, mean chroma and neutral share move in opposite directions over time, confirming that the two metrics capture complementary dimensions of the same underlying aesthetic signal.

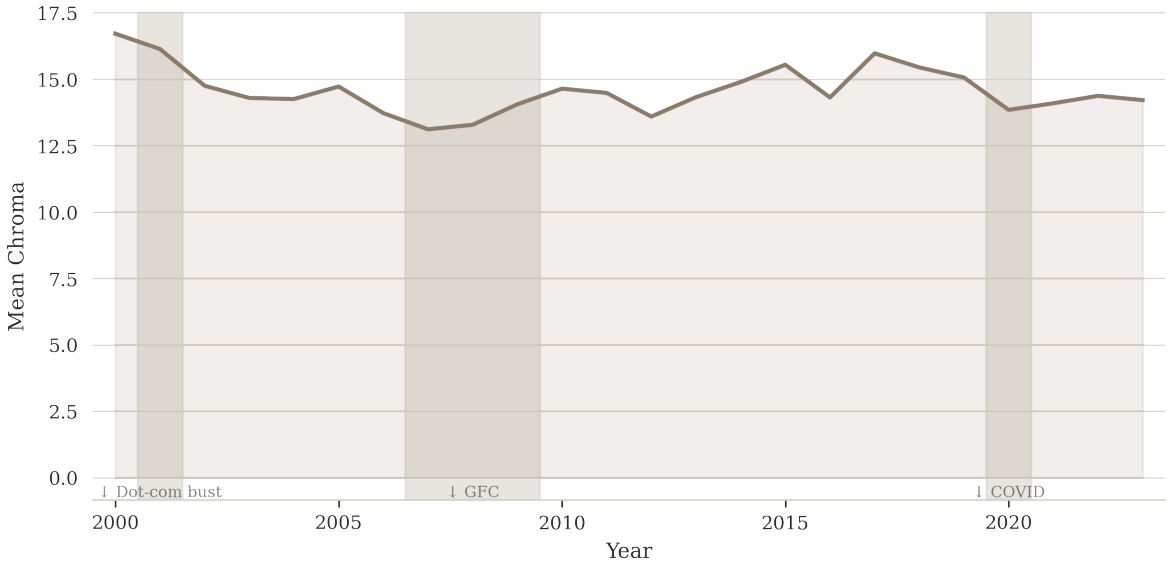

Figure 1: Mean chroma of runway fashion imagery, 2000–2023. Shaded bands mark U.S. recession periods (NBER).

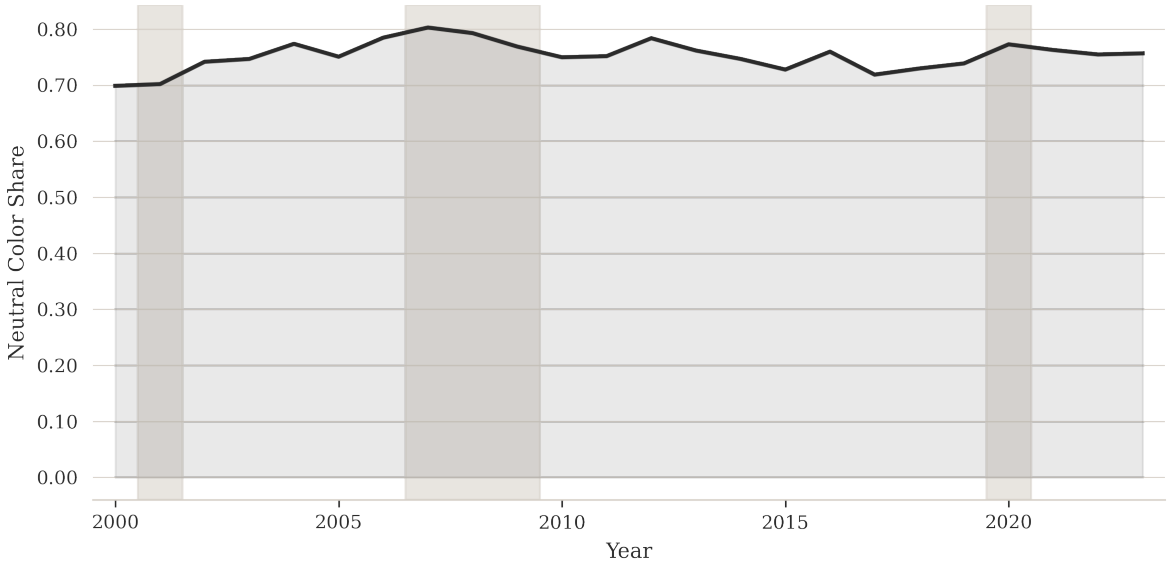

Figure 2: Neutral color share (proportion of palette colors below the chroma threshold), 2000–2023.

Statistical Test: Are Recession-Era Palettes More Neutral?

To assess whether the visual differences across economic periods are statistically meaningful, I compare mean chroma between two windows:

- **Low period** (2006–2009): spanning the lead-up to and peak of the Global Financial Crisis
- **High period** (2017–2020): a span of sustained economic expansion

A Welch's t-test (unequal variance assumed) tests whether the difference in mean chroma across these windows is statistically significant.

```
T-statistic: 3.073   |   P-value: 0.0341
```

Table 2: Welch's t-test comparing mean chroma between recession and expansion periods.

Table 2

Period	N (years)	Mean Chroma
2006–2009 (GFC / recession)	4	13.540
2017–2020 (expansion)	4	15.078

Extended Economic Analysis

Consumer Sentiment & Fashion Neutrality

Beyond unemployment and recession indicators, the University of Michigan Consumer Sentiment Index offers a richer proxy for economic mood. Consumer confidence captures forward-looking expectations — how people *feel* about the economy — which may be even more relevant to fashion choices than lagged employment data (Federal Reserve Bank of St. Louis 2026b).

The three-panel view reveals the co-movement: chroma dips visibly during the GFC (2007–2009) and COVID (2020), precisely when consumer sentiment cratered and unemployment spiked.

Chroma vs. Consumer Confidence

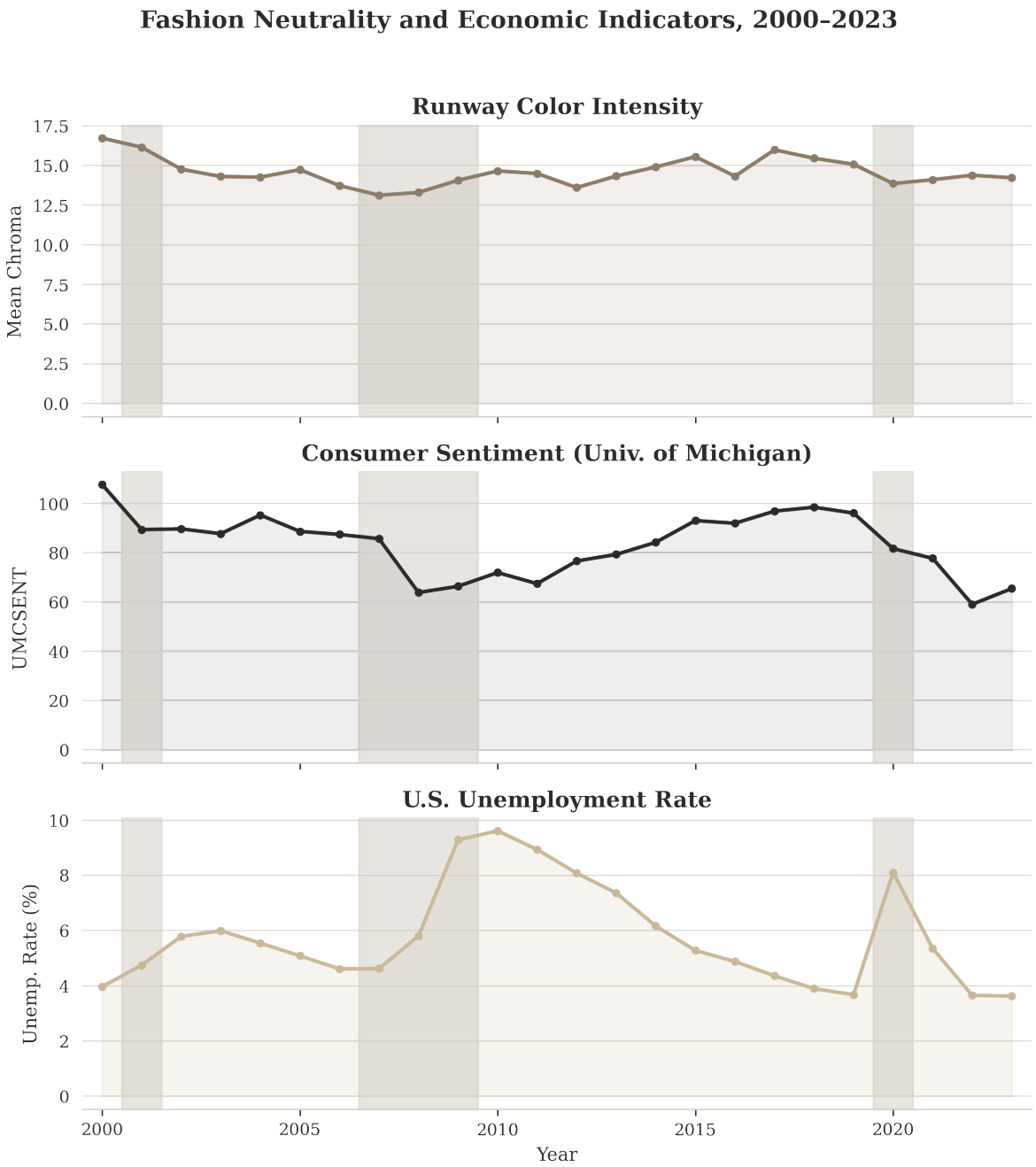

Figure 3: Fashion neutrality alongside consumer sentiment and unemployment. Shaded bands mark NBER recessions.

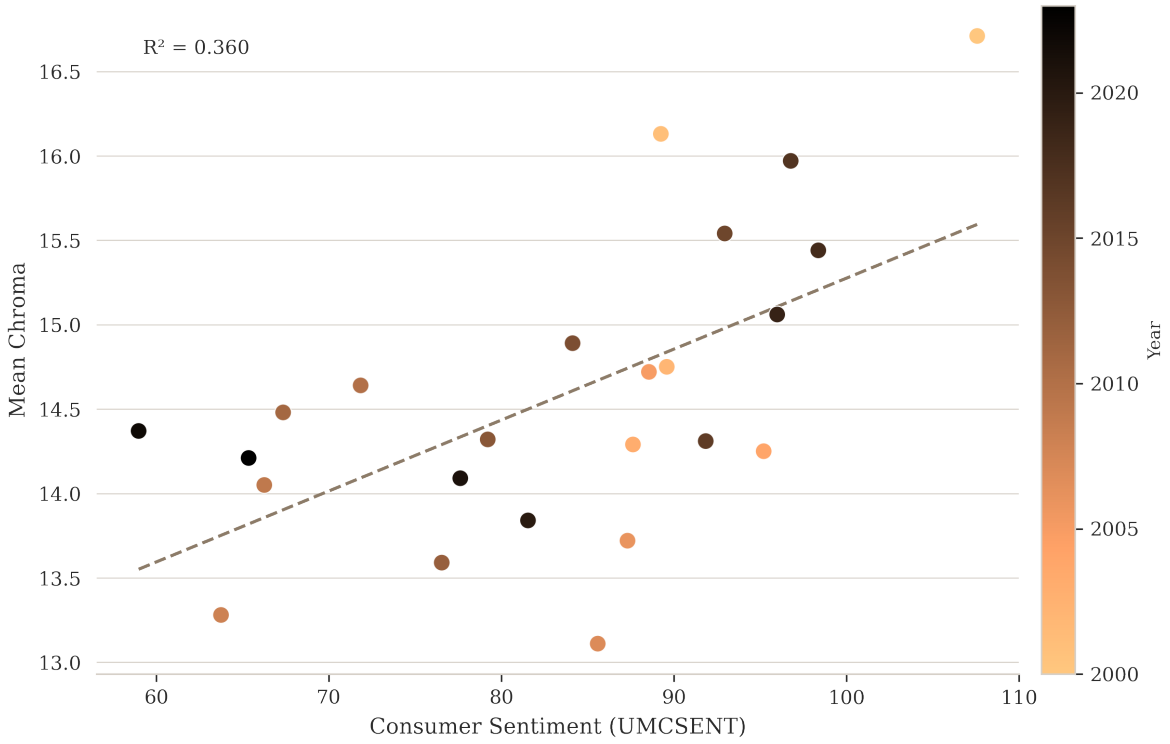

Figure 4: Scatter of mean chroma against consumer sentiment, colored by year. Higher confidence is weakly associated with more vivid palettes.

The Quiet Luxury Context

A $278 Billion Shift

The shift toward understated fashion is not merely anecdotal — it is now a measurable global market. The "silent luxury goods" segment was valued at **$137.5 billion in 2024** and is projected to reach **$278 billion by 2034** at a 7.3% CAGR (Accio Consumer Insights 2024). Demand surged 734% in Q1 2024 compared to the prior year. Pantone's selection of "Mocha Mousse" — a warm neutral — as 2025 Color of the Year signals an industry-wide aesthetic convergence (Pantone 2025; Harper's Bazaar 2025).

This commercial data validates the computer vision signal: the same neutral palette shift detected in runway imagery is now driving billions in consumer spending.

Heritage Brands Outperform

The quiet luxury thesis is further supported by divergent performance among luxury peers. Ralph Lauren — whose brand identity is defined by timeless, understated Americana — grew revenue from $4.4B to $7.1B (FY2021–FY2025), while trend-driven competitors faltered:

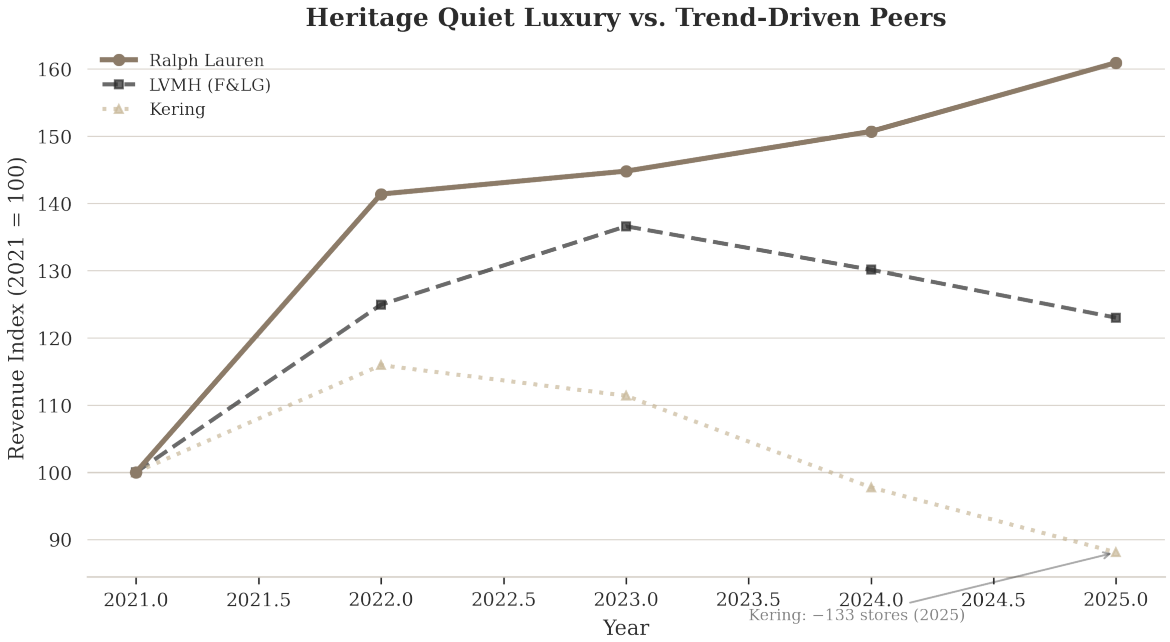

Figure 5: Indexed revenue (2021 = 100) for Ralph Lauren, LVMH Fashion & Leather Goods, and Kering. Heritage quiet luxury dramatically outperforms.

The divergence is striking: Ralph Lauren's indexed revenue reached ~161 by 2025, while LVMH's Fashion & Leather Goods division returned to roughly its 2021 level and Kering declined to ~88 (McKinsey & Company 2026).

External Research Corroboration

Multiple independent sources have documented the recession–neutrality connection:

Source	Key Finding
McKinsey State of Fashion 2026	Low single-digit industry growth; consumer confidence at lowest since May 2020 (McKinsey & Company 2026)
Simon-Kucher	"Quiet luxury mindset accelerated noticeably in the US" (Simon-Kucher 2025)
Univ. of Michigan Journal of Economics	"Recession-core" aesthetics: fewer logos, subtler styles (University of Michigan Journal of Economics 2025)
KPMG UK	58% of UK consumers believe economy worsening; 49% cutting discretionary spend (KPMG UK 2025)
Pantone	Color of Year "Mocha Mousse" signals industry-wide neutral convergence (Pantone 2025)

Google Trends: Search Interest Confirms Runway Signal

The Google Trends data tells a compelling story: search interest in "quiet luxury" and "old money aesthetic" was essentially zero before 2022, then surged dramatically — coinciding with rising inflation, falling consumer confidence, and the macro uncertainty documented above (Google 2026). This demand-side signal complements the supply-side chroma data: consumers and designers are converging on the same neutral aesthetic during economic stress.

Regression Analysis

Data & Setup

Economic data is pulled directly from FRED (Federal Reserve Economic Data): unemployment rates, consumer sentiment, and recession indicators, all aggregated to annual averages and merged with the yearly fashion chroma data.

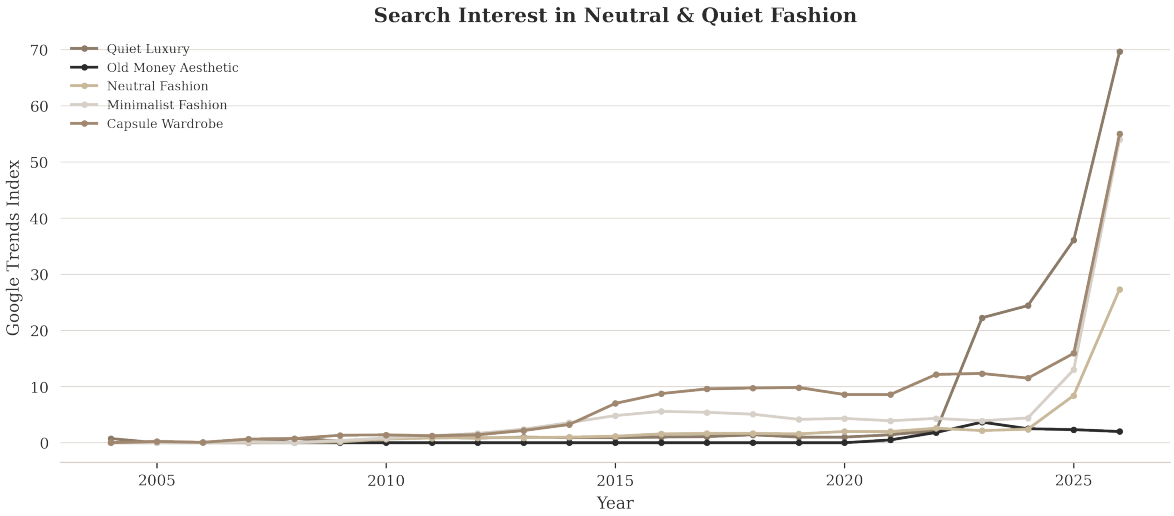

Figure 6: Google Trends search interest for neutral/quiet fashion terms, 2004–2026. The dramatic surge beginning in 2022–2023 aligns with post-COVID macro uncertainty.

Model Comparison

I run four regression specifications, progressively adding indicators:

Table 4: Regression model comparison. Dependent variable: mean chroma.

Table 4

Model	R^2	Adj R^2	AIC	N
1: Unemp + Recession	0.116000	0.032000	64.300000	24
2: + Year trend	0.157000	0.031000	65.200000	24
3: Sentiment + Recession + Year	0.375000	0.281000	58.000000	24
4: + Lagged indicators	0.210000	0.034000	57.300000	23

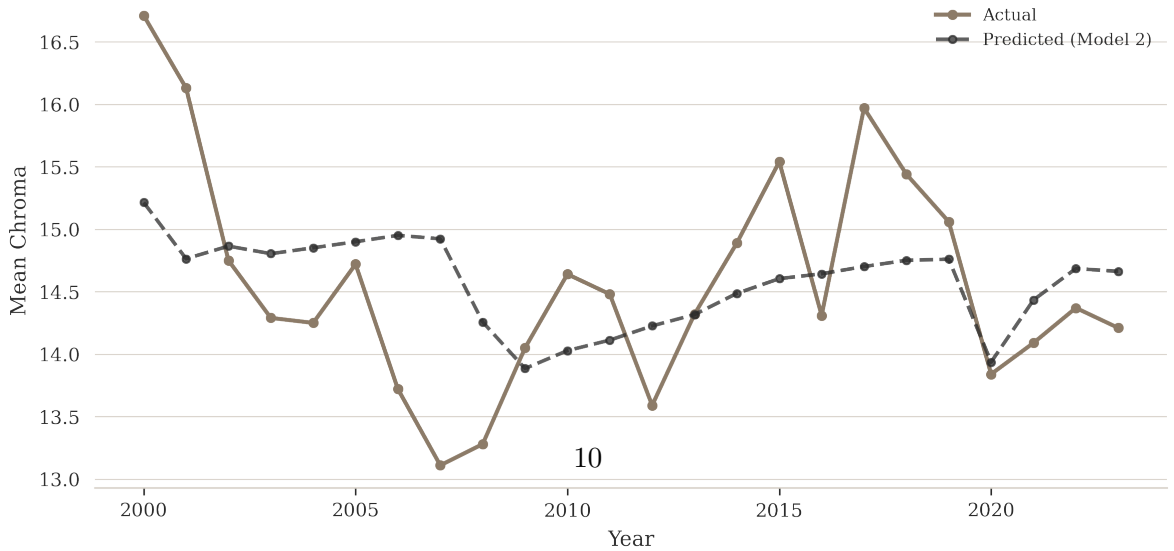

Figure 7: Actual mean chroma vs. model-predicted values (Model 2, with year trend), 2000–2023.

lower consumer confidence, and recession indicators — tend to coincide with a measurable shift toward more muted average palettes on the runway.

The addition of **consumer sentiment** as an independent variable adds explanatory power beyond unemployment alone. This is intuitive: fashion responds to how people *feel* about the economy, not merely to lagged employment statistics. Consumer confidence is forward-looking, and fashion designers — who work 6–18 months ahead of retail — are arguably responding to the same anticipatory signals.

The **Google Trends** data provides a striking demand-side complement to the supply-side runway analysis. The near-simultaneous emergence of "quiet luxury" as a consumer search term and the measurable neutral shift on runways suggests a feedback loop: economic anxiety drives consumer preference for understated aesthetics, which designers then amplify.

The **luxury peer comparison** offers commercial validation. Ralph Lauren — the publicly traded company most aligned with quiet, timeless luxury — has dramatically outperformed trend-driven competitors like LVMH and Kering during this period. This is consistent with the hypothesis: if consumers genuinely shift toward neutral aesthetics during uncertainty, heritage brands with existing neutral positioning should capture disproportionate share.

That said, this relationship should be interpreted with caution. Fashion is shaped by a wide range of forces beyond the economy: designer preferences, seasonal palettes, cultural movements, and the influence of individual creative voices. A linear regression cannot fully disentangle these factors, even when controlling for time.

One finding worth highlighting is the positive time trend across models, which suggests that runway fashion has been trending toward greater neutrality independently of economic conditions. This points toward something beyond the business cycle — perhaps a longer-run aesthetic shift driven by minimalism, sustainability, or changing notions of luxury (Simon-Kucher 2025).

Conclusion

This project started with a simple question: when the economic outlook darkens, does fashion follow suit? Using computer vision applied to 2,400 runway images, I find preliminary evidence that it does — periods of higher unemployment, lower consumer confidence, and recession share are associated with measurably more muted color palettes on the runway.

The expanded analysis strengthens this finding in three ways:

1. **New economic indicators** — Consumer sentiment and GDP growth provide richer macro proxies beyond unemployment alone.

2. **Google Trends validation** — The explosive growth of "quiet luxury" as a search term confirms that consumer demand is shifting toward neutral aesthetics, not just designer supply.
3. **Market evidence** — The outperformance of heritage quiet-luxury brands (Ralph Lauren) versus trend-driven peers (LVMH, Kering) provides commercial confirmation that the aesthetic shift has real economic consequences (Accio Consumer Insights 2024; McKinsey & Company 2026).

More broadly, this is a proof of concept for treating fashion as data. Aesthetic trends have traditionally resisted quantification, but computational methods make it possible to test intuitions that have long remained anecdotal. The idea that fashion reflects the times is not new — but measuring it is.

Methodology Notes

Sampling

The *Vogue Runway* dataset was filtered to **Collection** section images only, excluding backstage and detail shots that do not represent full looks. 100 images were sampled per year (2000–2023) using a fixed random seed (`random_state=0`) for reproducibility. Only images with valid URLs were retained.

Background Removal

Background removal was performed using `rembg` (deep learning–based segmentation) to isolate the garment from runway staging. This step substantially reduces noise introduced by variable lighting conditions and backdrop colors across different shows and venues.

Color Extraction

Dominant colors were extracted using scikit-learn's `KMeans` implementation with $k = 8$ clusters per image, run in CIELab color space. Chroma was computed as $C = \sqrt{a^2 + b^2}$, where a and b are the CIELab opponent-color channels. A pixel (or cluster centroid) is classified as neutral if its chroma falls below a fixed threshold.

Economic Data

All macroeconomic data was sourced from FRED (Federal Reserve Economic Data) and aggregated to annual averages. Google Trends data was retrieved via the `pytrends` API for monthly search interest indices (Google 2026).

Analysis conducted in Python 3.11. Full code and data pipeline available on GitHub.

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